Medical Affairs, work redesigned



AI in Action · October 13–14, 2026
Convene Philadelphia

Vivek Mukhatyar

DISCLOSURE

The views presented here are my own and do not reflect those of my current employer.

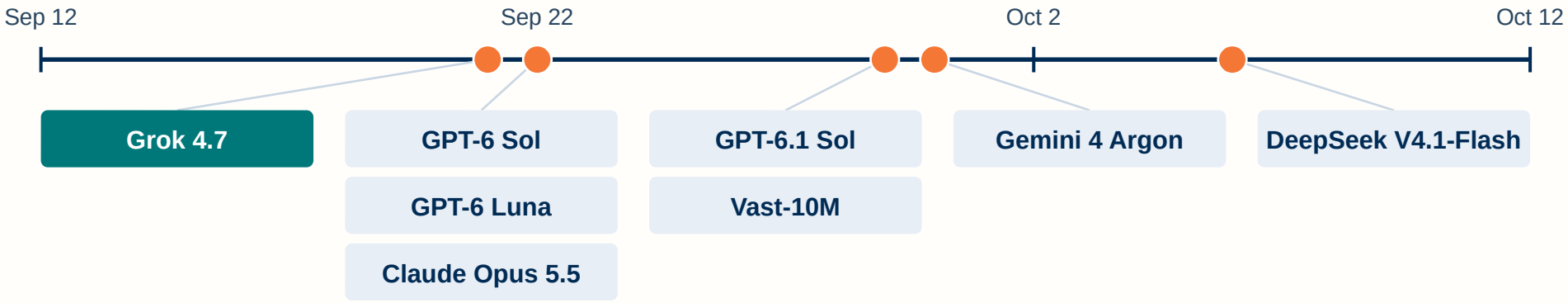
Vivek Mukhatyar

The last 30 days in AI

September 12 – October 12, 2026

8

frontier model releases
from 6 labs in 30 days



MATH

Navier–Stokes

machine-checked claimed proof

Produced by AI agents (Sep 8), under expert review

+ 722 math papers released

COST OF INTELLIGENCE

$4

per million tokens, median frontier

−49% in 30 days

−84% vs March 2023

BIOLOGY

37,000

AI agents across ~50,000 trials

Stanford Virtual Biotech (Sep 17)

Claude finds a new CRISPR-like enzyme (Sep 23)

SCALE OF MEMORY

10M

tokens in a single prompt

Vast-10M: first 10-million-token context window

≈ a library of trial reports in one prompt

Sources (Sep 12 – Oct 12, 2026): vendor announcements (xAI, OpenAI, Anthropic, Voltropy, Google DeepMind, DeepSeek); OpenAI Navier–Stokes post, independent Lean rebuild, New Scientist; PicksByModel Token Price Index; BenchLM; Stanford Medicine. Some figures vendor-reported.

Agenda: six acts, then we build

ACT 1

1

From answers to work

ACT 2

2

The interface is changing

ACT 3

3

Context engineering

ACT 4

4

From chatbot to agent swarm

ACT 5

5

My use case: launch planning

ACT 6

6

Your turn

Grok Bot, the library, the build

THEN, HANDS-ON

11:15 AM
Team creation

1:45 PM
Build the AI worker

Wed 9:00 AM
Demo day

Four objectives for today

01

Give AI goals, not questions

Move from summarizing and finding answers to orchestrating and executing whole workflows



02

Meet the frontier agents

Grok Bot, Claude.ai, Microsoft Copilot, ChatGPT, Meta Muse: your own personal workforce



03

Context is our secret sauce

Prompt engineering becomes context engineering. Medical Affairs holds the context.



04

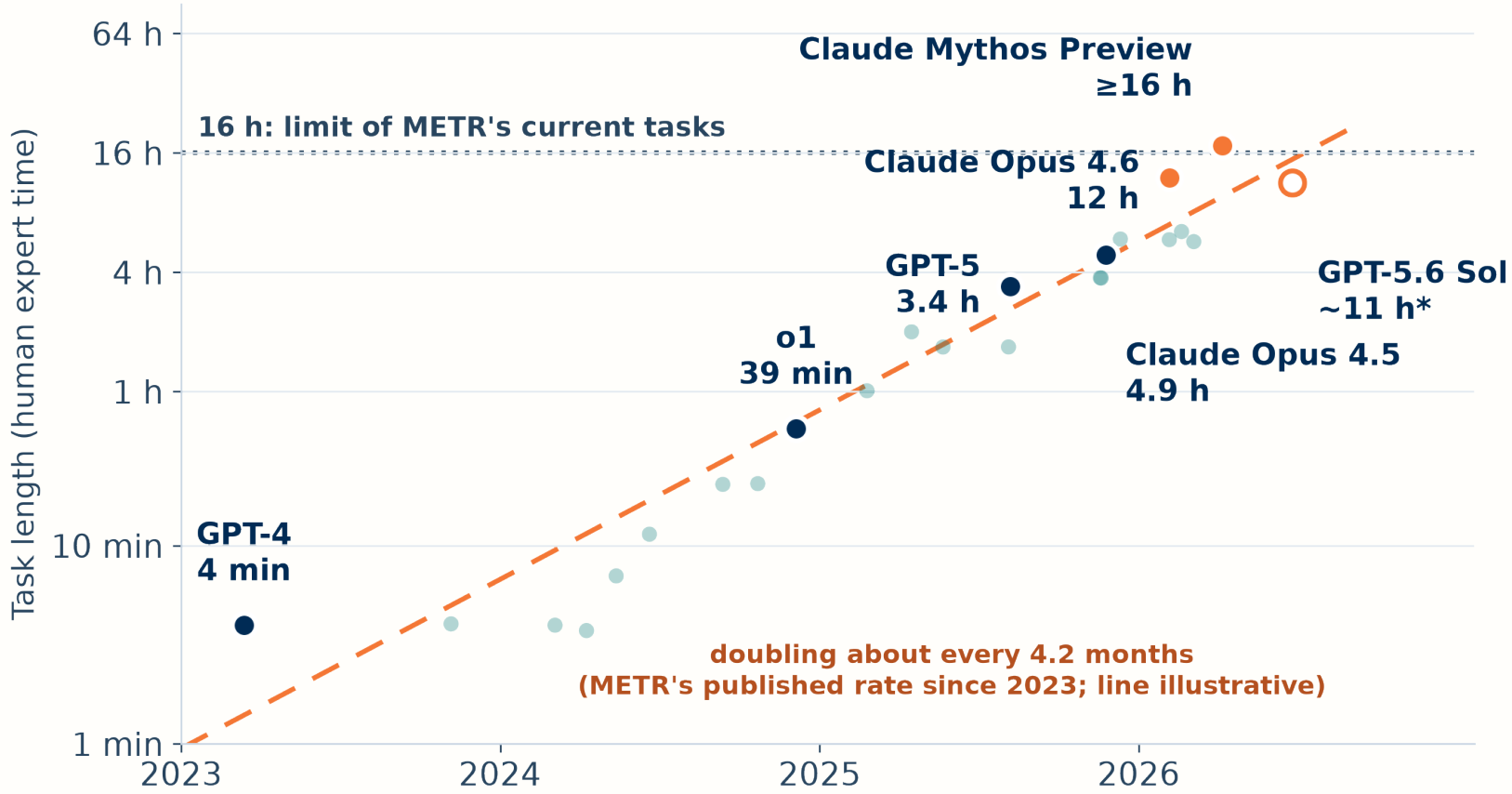
Have fun building

Form a team, build a real AI worker together, and show it on demo day



AI agents went from minutes-long tasks to day-long tasks in three years

METR 50% time horizon: about 4 min for GPT-4 (March 2023) → 12 h for Claude Opus 4.6 and at least 16 h for Claude Mythos Preview (2026)



SEPTEMBER 2026 FRONTIER
too new for METR

Epoch Capabilities Index (ECI)
Epoch AI composite of many benchmarks, in index points (not hours)

Claude Opus 5.5	167.3
GPT-6 Astra	166.5
Grok 4.7	153.5

Task length AI can complete has doubled about every 4 months since 2023, and the newest models have outgrown the test.

METR time-horizons (updated May 2026) and METR posts on X; Epoch AI Capabilities Index. 50% success on well-defined software tasks, measured by how long a human expert takes.
Doubling: METR 128.7 days since 2023 (95% CI 104–158). Top points have wide ranges (Opus 4.6: 5–61 h; Mythos: 8–55 h). *GPT-5.6 Sol: pre-deployment, METR post on X.

Scientific questions already reach AI at scale

300M+

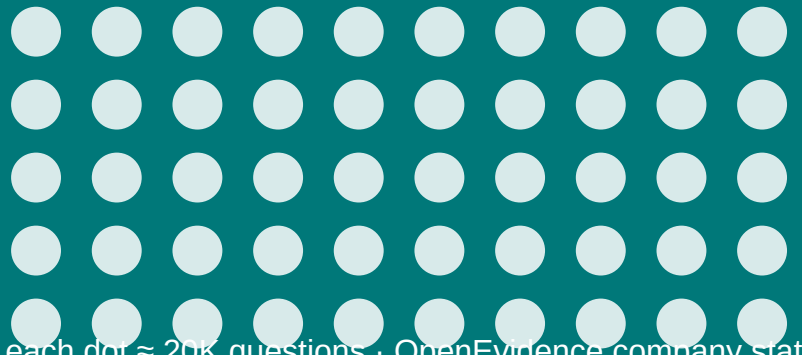
people a week ask ChatGPT health questions



each dot ≈ 6M people · OpenAI, July 2026

1M+

questions a day answered by OpenEvidence



each dot ≈ 20K questions · OpenEvidence company statement, 2026

Different populations and units. Both change where scientific exchange begins.

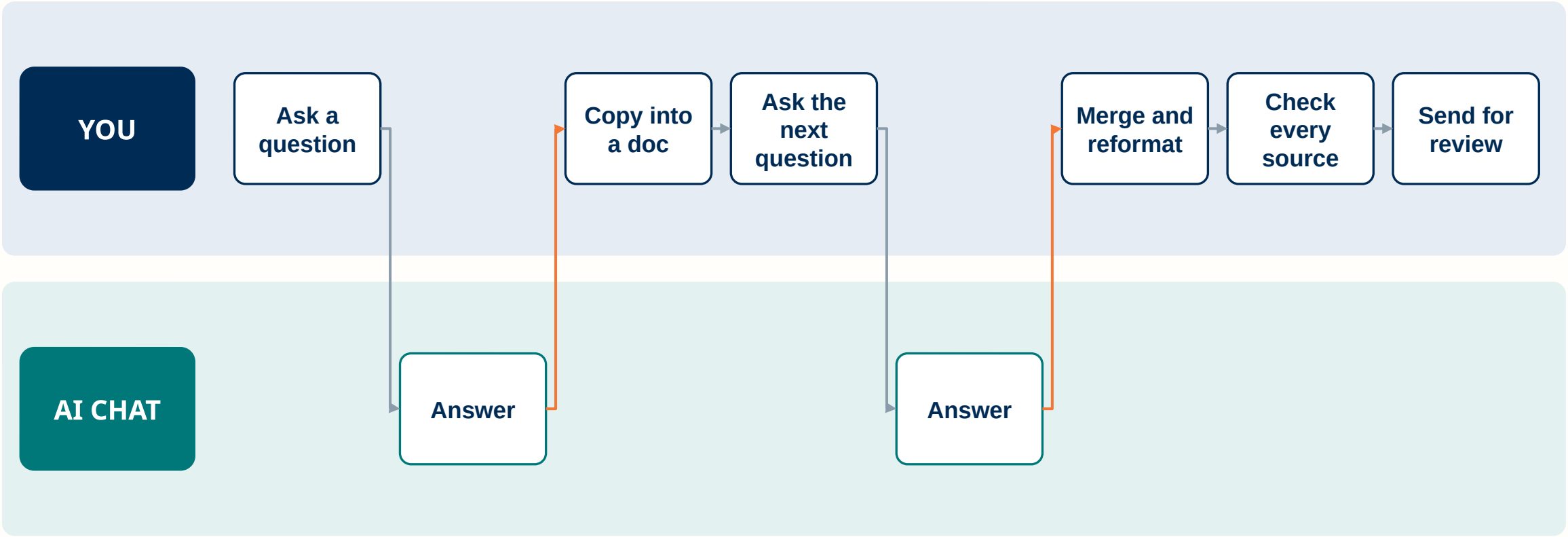
The north star remains the same

Patient outcomes

via evidence · exchange · insights



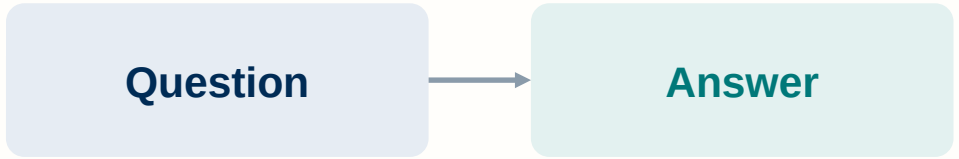
Most of us start here: one answer at a time



The human carries every handoff. The orange arrows are you, picking the work back up.

The problem is not chat. It is the unit of work.

CHAT



One exchange. Then you start over.

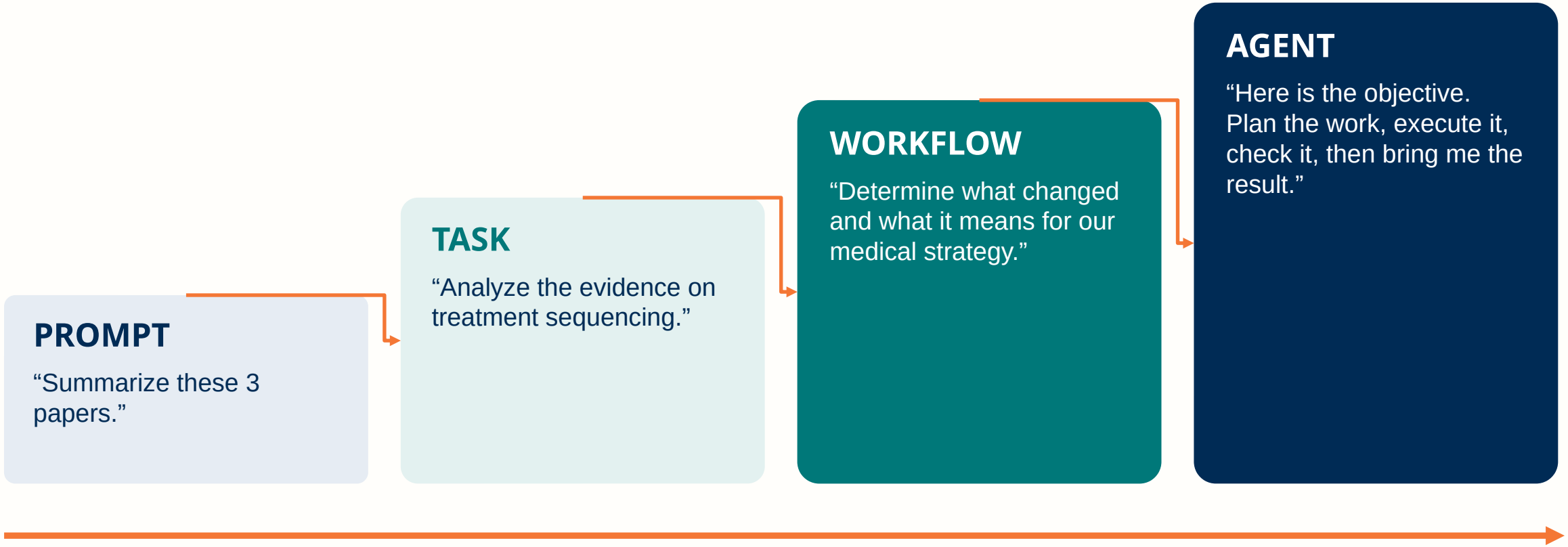
WORKFLOW



Many steps, one outcome, checked before it reaches you

Stop optimizing the answer. Start designing the work.

Prompt → task → workflow → agent



From prompts to assignments

A PROMPT

> “Summarize obesity papers”

One answer. Repeated handoffs.



AN ASSIGNMENT

Goal	Reviewable ad-board package: sustaining treatment
Context	Our evidence position, house rules, past questions
Sources	PubMed, ClinicalTrials.gov, approved inputs
Output	Charter, agenda, pre-read, claim ledger
Checks	Every claim cited; limitations stated
Review	A named human approves before use

Stop asking AI questions.

Start giving it jobs.

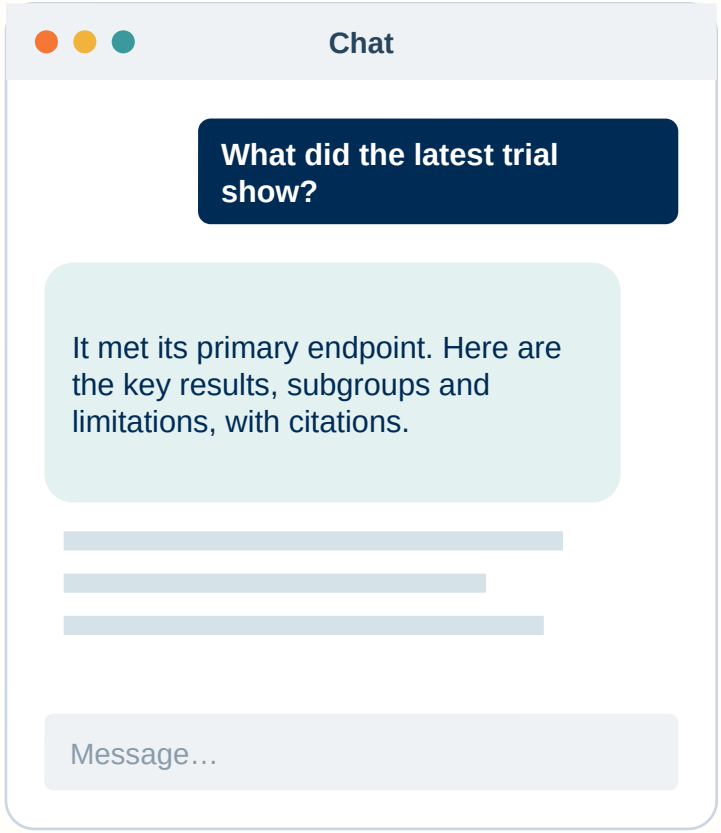
ACT 2

The interface is changing

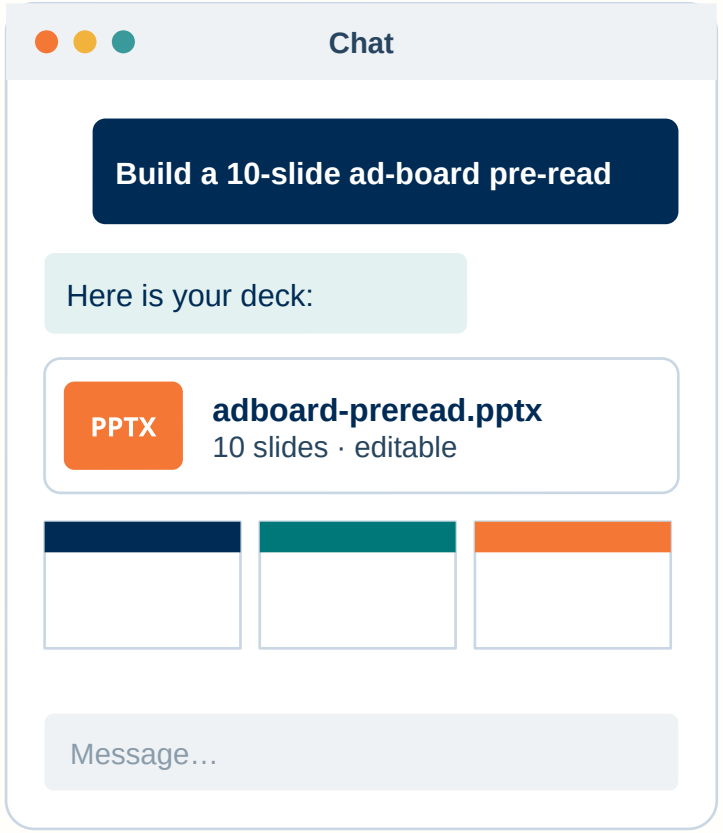


The interface changed: same window, different job

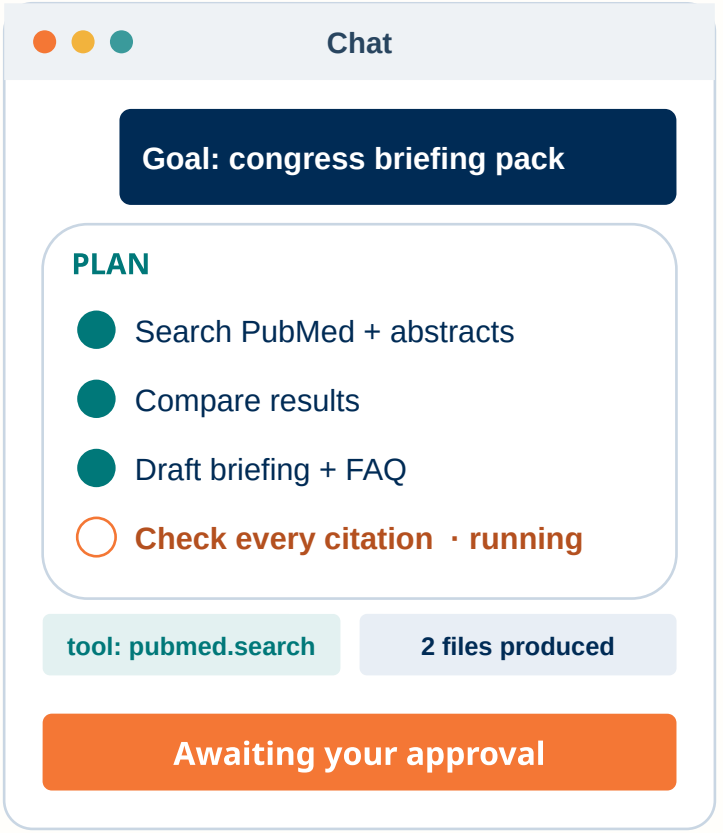
1 · ANSWERS



2 · ARTIFACTS



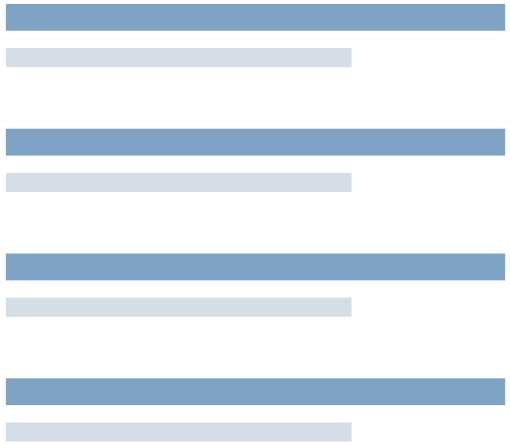
3 · AGENTS DOING WORK



Answers → artifacts → agents doing work. You move from asking to approving.

From finding information to completing work

2022 · SEARCH



Find pages

2024–25 · ANSWER ENGINES

Synthesized answer
One response, built from many sources

[1] [2] [3]

Synthesize responses

2026 · WORKFLOW EXECUTION

Plan
↓
Act
↓
Inspect
↓
Deliver

Plan · act · inspect · deliver

Meet the frontier agents

USED TODAY

Grok Bot

xAI

Claude.ai

Anthropic

Microsoft Copilot

Microsoft

ChatGPT

OpenAI

Meta Muse

Meta

CHAT TOOLS
You ask, it answers

the shift

PERSONAL AGENTS
Your own workforce

Same assignment and checks · different access and controls · use what your workplace provides

Your own workforce: a small team, clear handoffs

YOU · set the goal, approve the result

Coordinator	Owns the outcome
Evidence worker	Returns sources + limitations
Design worker	Returns decision map + draft
Reviewer	Returns defects + severity



Every handoff carries inputs, sources, open questions and a done-check

The new vocabulary of AI

Model

The reasoning engine that reads and writes language

Agent

A model pursuing a goal through tools and feedback

Harness

The environment that manages tools, state, permissions and checks

Context

Everything the model can see for this step

Memory

Saved information and progress across steps

Skills

Reusable instructions for a type of work

Tools

Permitted ways to reach a source or take an action

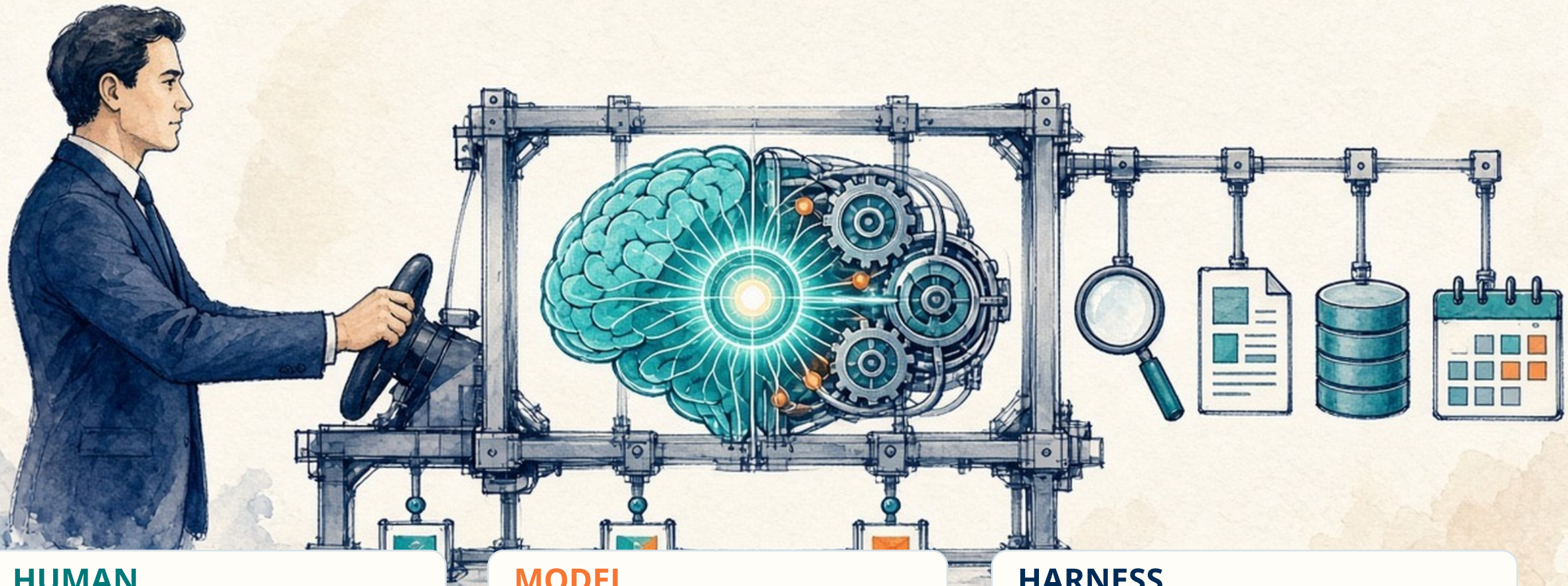
Swarm

Several specialists coordinated around one outcome

Long horizon

Work that runs across many steps and hours

What is an AI agent?



HUMAN

Purpose · boundaries · review

MODEL

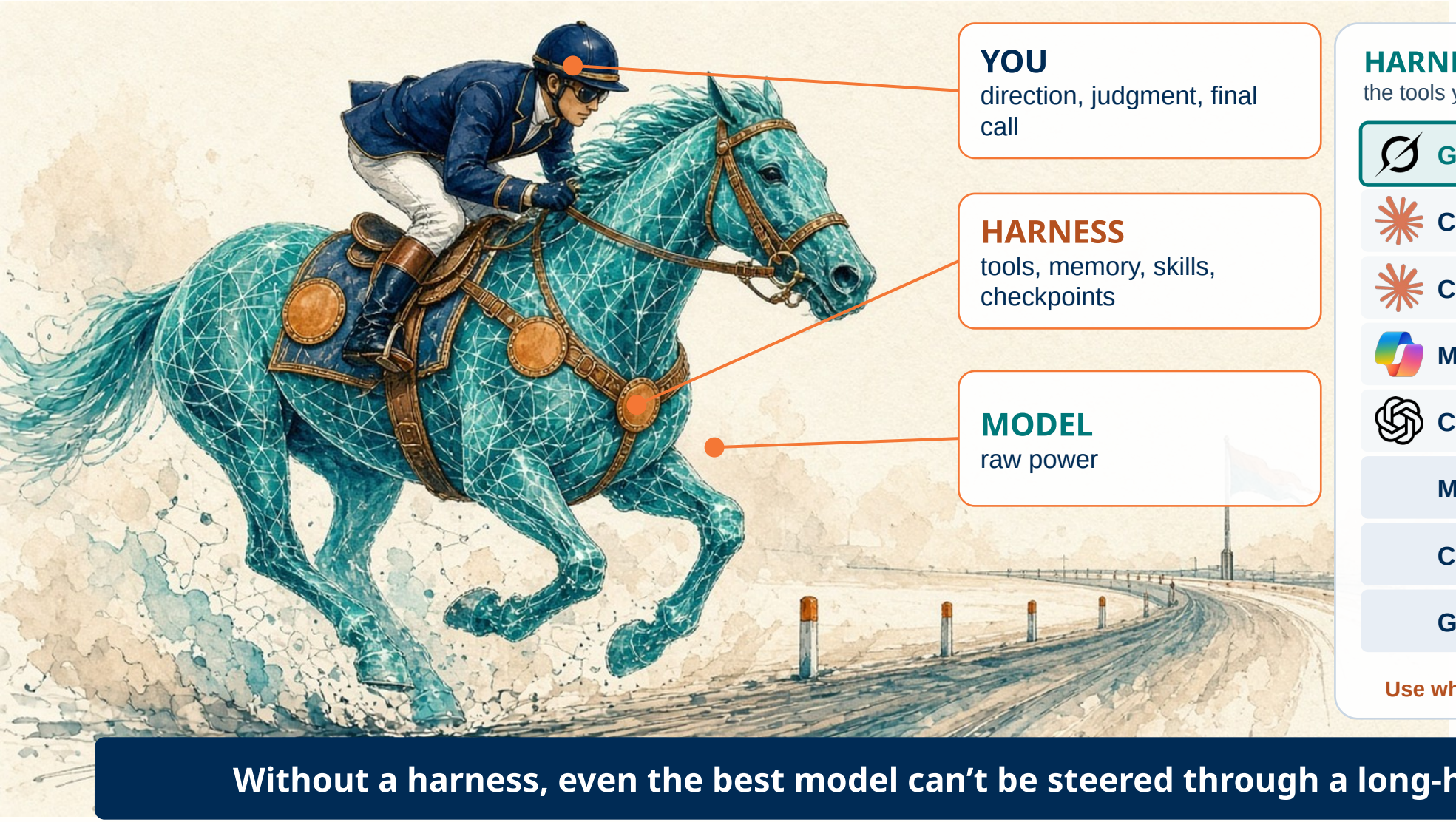
Reasoning + language

HARNESS

Tools · memory · permissions · checks

An agent = a model + a harness + human direction, working on a real job

The model is the horse. You are the jockey. The harness is the tool.



YOU
direction, judgment, final call

HARNESS
tools, memory, skills, checkpoints

MODEL
raw power

HARNESSES
the tools you steer the model with

- Grok Bot — USED TODAY
- Claude Code
- Claude Cowork
- Microsoft Copilot
- ChatGPT (agent mode)
- Meta Muse
- Cursor
- Gemini

Use what your workplace provides.

Without a harness, even the best model can’t be steered through a long-horizon task.

Context is the secret sauce, and Medical Affairs holds it



Prompt

How you ask

Instructions + output format

Context

What it can see

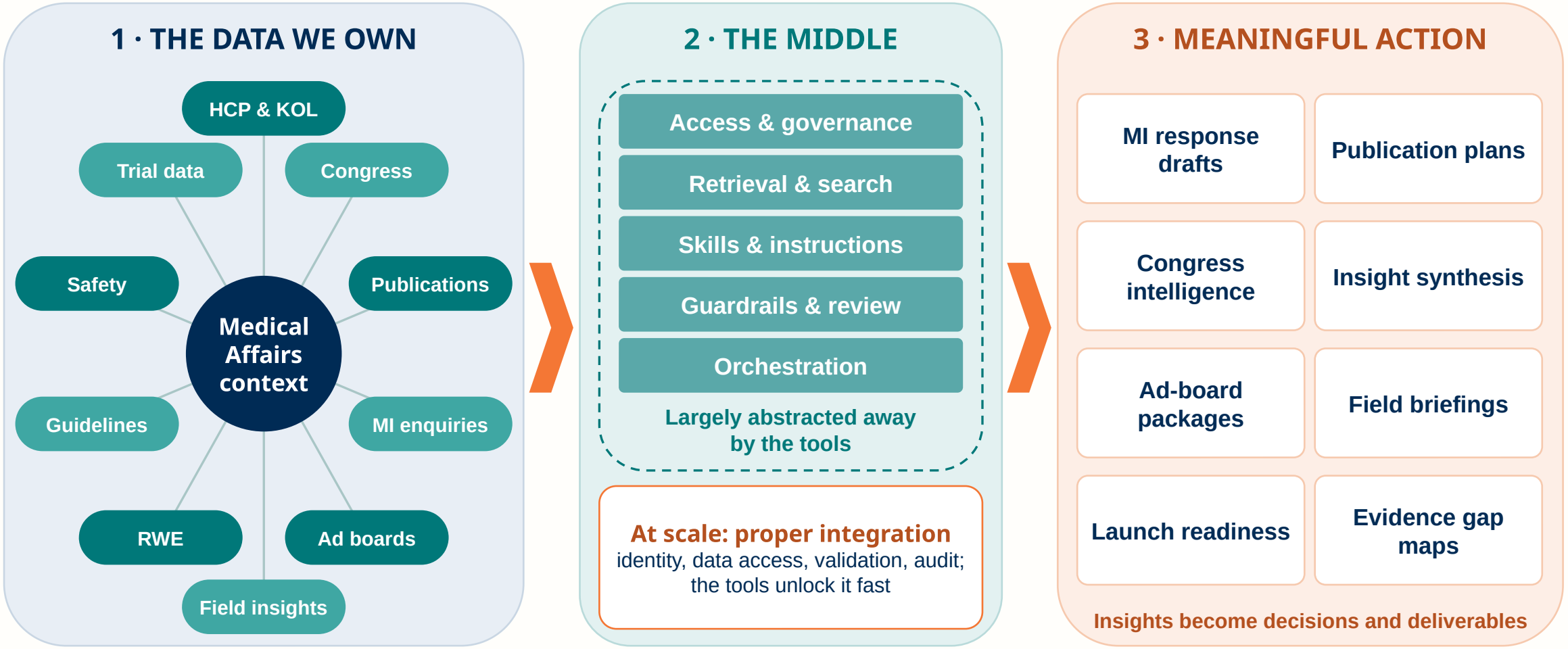
Held by Medical Affairs: evidence, house rules, past decisions, field insight

Harness

How the work runs

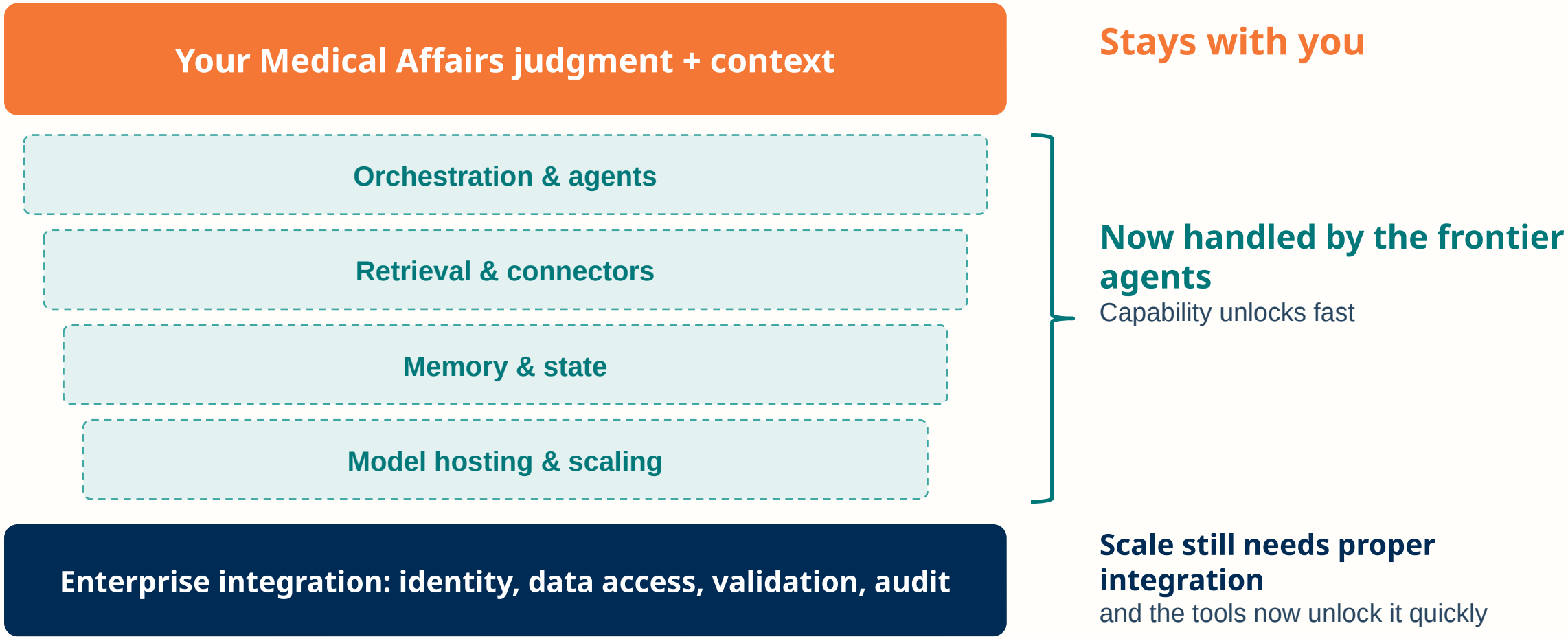
Tools + permissions + checkpoints

From the data we own to meaningful action

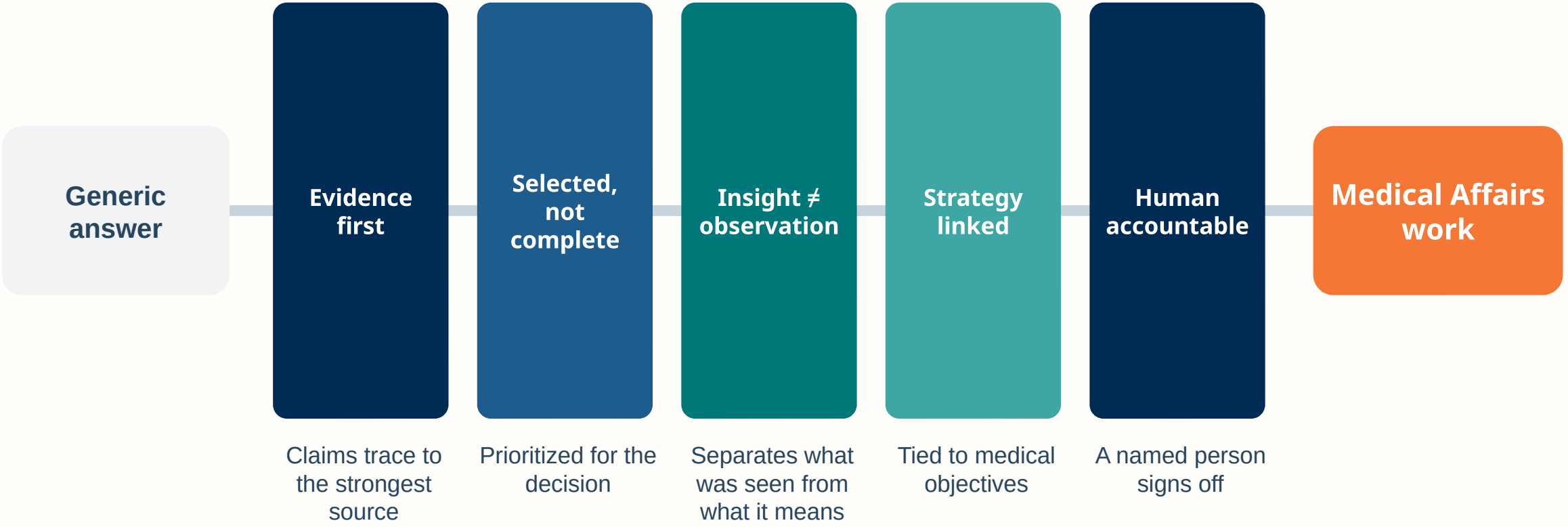


We own both ends: the context going in and the judgment on what comes out.

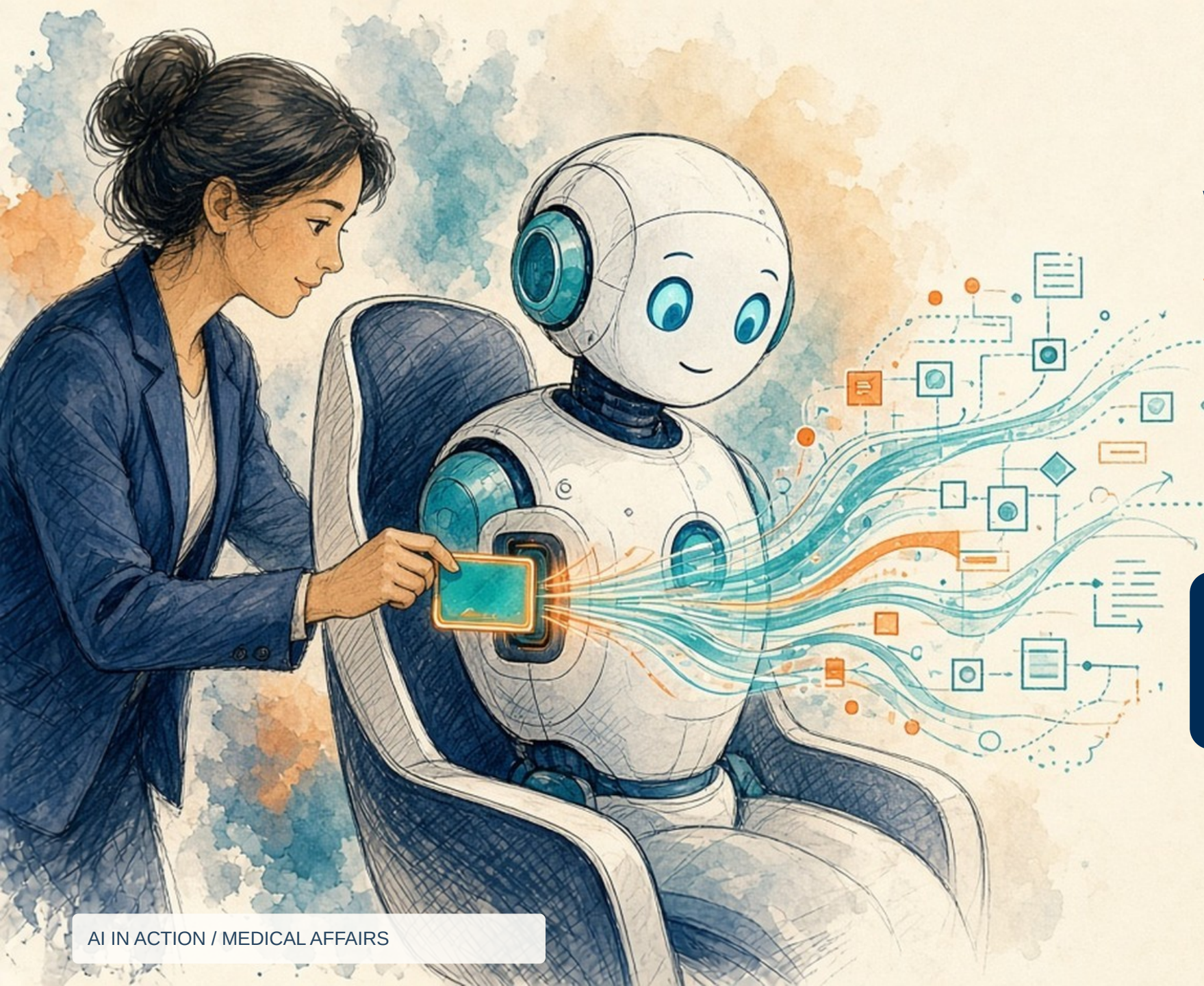
The middle is being abstracted away



What makes it Medical Affairs: encoded judgment



You are not just using AI. You are teaching it how your function thinks.



What is a skill?

“I know Kung Fu.”

Load the skill.
Use it instantly.

Watch the clip
The Matrix 25th Anniversary · I Know Kung Fu · Warner Bros. (YouTube)

Link only: youtu.be/ozS_6vqnJ8w

A skill is a recipe card for your agent

Written once, kept on the shelf, read only when the task calls for it.

	IN THE KITCHEN a recipe card	IN MEDICAL AFFAIRS an SOP	FOR AN AGENT a skill
WHEN TO USE IT	Sunday dinner for six	An MI enquiry arrives	`description: "Use when…"`
WHAT IT NEEDS	Chicken, thyme, 90 minutes	Label, standard responses	**Templates, reference files, scripts**
THE STEPS	Season, roast, rest	Log, search, draft, review	**Step-by-step instructions in SKILL.md**

HOW YOUR AGENT PICKS ONE UP

1. **Sees only each skill's name and description**
2. **Your request matches one description**
3. **Reads that one SKILL.md in full**
4. **Follows the steps, opens extras only if needed**

Inside a skill: one folder, one SKILL.md

FOLDER

msl-pre-call-planning/

SKILL.md
required · the instructions

scripts/
optional · code it can run (ma_render.py)

templates/
optional · forms and layouts to fill

reference/
optional · house rules, worked examples

SKILL.md

```
name: msl-pre-call-planning

description:
Prepare an MSL for an upcoming scientific HCP
engagement… Use for pre-call briefs, meeting
objectives, anticipated scientific questions…

# MSL Pre-call Planning
## Collect once
## Prepare the brief
- Why this meeting  - What changed
- Three questions  - Likely questions
```

NAME
what it is called

WHEN TO USE
how the agent knows to pick it up

STEPS
what to do, in order

Real skill from the open Medical Affairs library, abridged

Skills for everyday Medical Affairs work

Each one is a folder in the open library: the same method, every time, for anyone on the team.

MSL pre-call brief

`msl-pre-call-planning`

One-page brief, evidence pack, open questions

Medical information response

`medical-information-response`

Draft reply to an unsolicited enquiry

Congress readout

`congress-intelligence`

What changed, why it matters, what to do

Abstract draft

`congress-abstract-and-poster`

Fits the congress rules, then the poster

Insights from field notes

`field-insight-synthesis`

Ranked insights with evidence and an owner

Advisory board

`advisory-board-design`

Charter, questions, and a usable summary

Works in Grok Bot (used today) and in other agents that read SKILL.md. Use what your workplace provides.

Agent Skills make expertise reusable

GitHub is the shared shelf
A skill is the work recipe

SKILL.md → procedure

Supporting files → examples

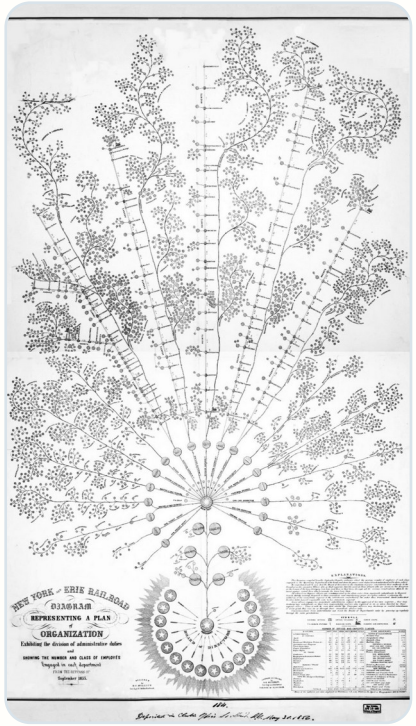
House rules → boundaries



170 years of the same org chart

Invented in the 1850s, barely changed since. AI agents will change it.

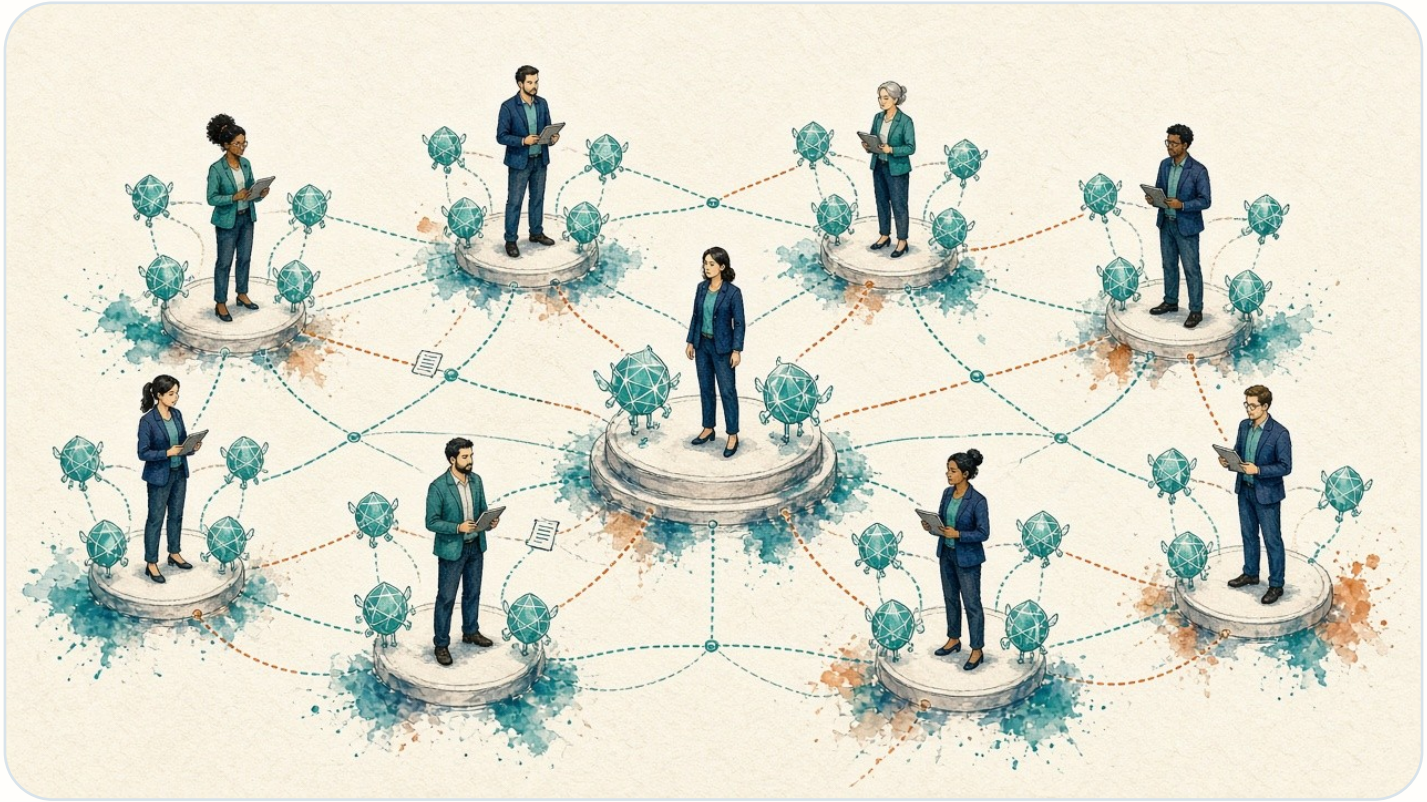
1 1855: the first org chart



Daniel McCallum, New York & Erie Railroad, 1855
(Library of Congress)

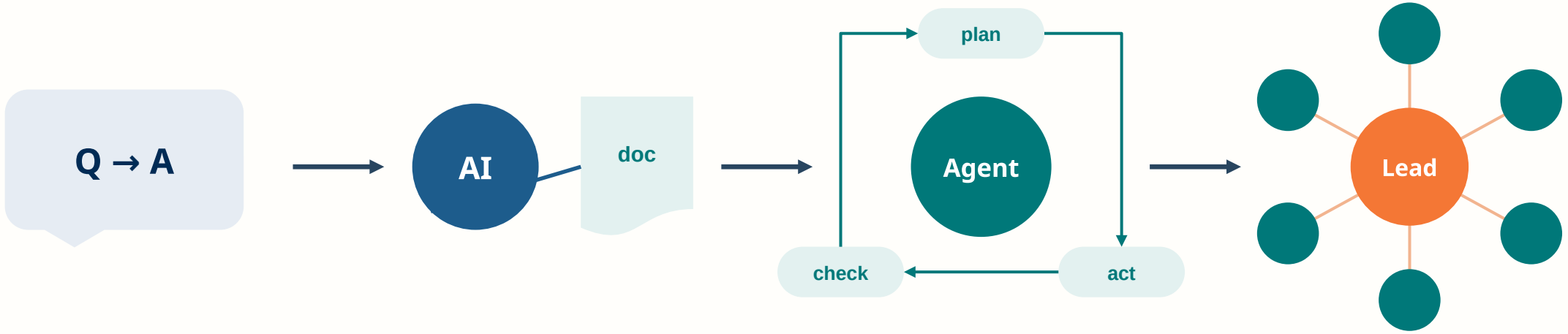
Tree-shaped. Still how we draw teams today.

2 Now: each person leads their own team of agents



Organized around outcomes, not boxes on a chart. Leading firms are already making this shift.

From chatbot to agent swarm



CHATBOT

Give: A question

Get: An answer

You do every step

ASSISTANT

Give: A task + some files

Get: A draft of one deliverable

You stitch drafts together

AGENT

Give: A goal + skills + house rules

Get: A finished workflow with provenance

You review and decide

AGENT SWARM

Give: A goal + an org chart + a context packet per role

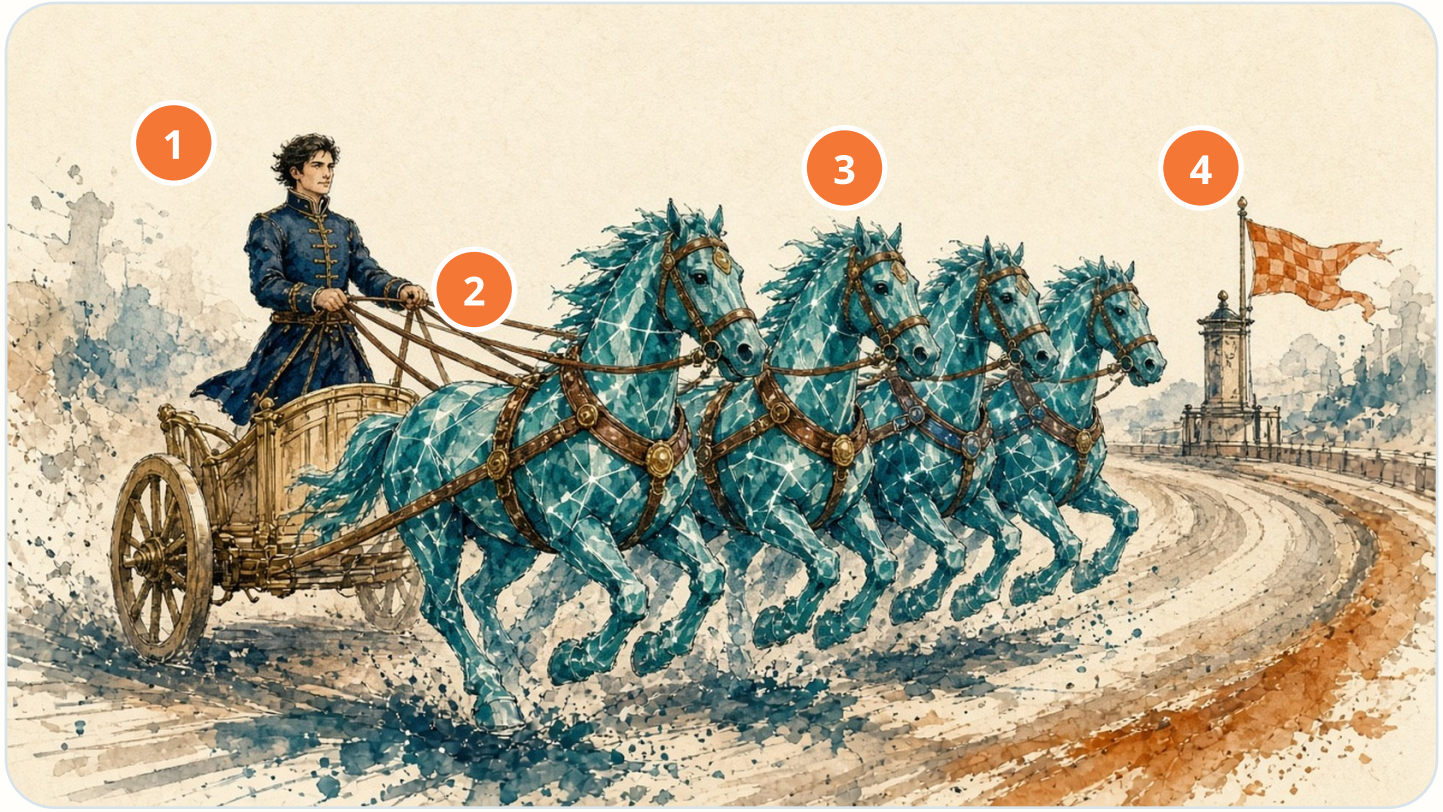
Get: A coordinated plan, an auditor, human gates

You lead and judge

The jump to a swarm is structure, not a bigger prompt: roles, context, handoffs, gates

What is an agent swarm?

A team of AI agents, each with its own role, skills and context. A lead agent splits the job, hands work between them, merges the results and stops at human checkpoints.



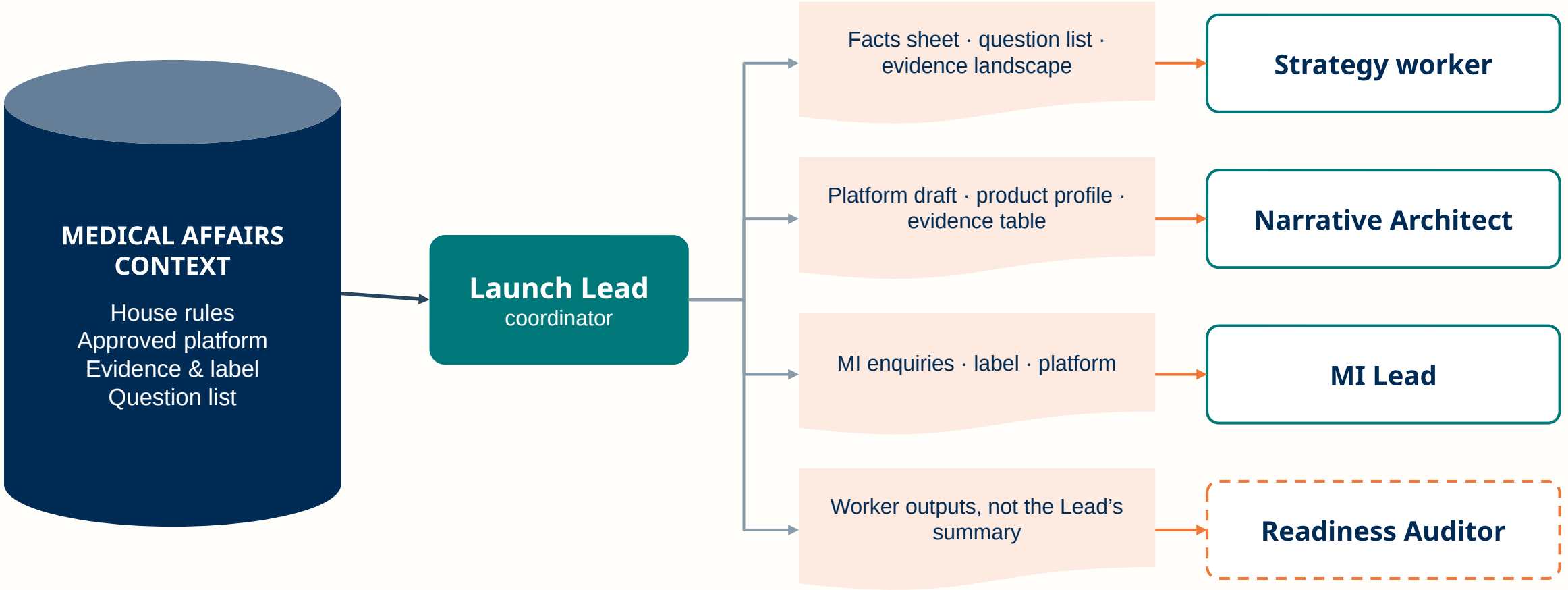
ONE HUMAN, MANY HORSES

1. **Charioteer = you**
   the human at the helm sets the direction
2. **Reins + harness = lead agent + rules**
   splits the job and hands work along
3. **Each horse = a specialist agent**
   its own skill and context
4. **Finish post = human checkpoint**
   nothing crosses without your decision

MODEL the horse → **HARNESS** tools + rules → **AGENT** + a job → **SWARM** many horses

One agent for a single task. **A swarm** for multi-part work with hand-offs, like a launch plan.

Context engineering feeds every sub-agent



1 Medical Affairs holds the context

2 Curate one source of truth

3 Slice a packet per role

4 Contracts at every handoff

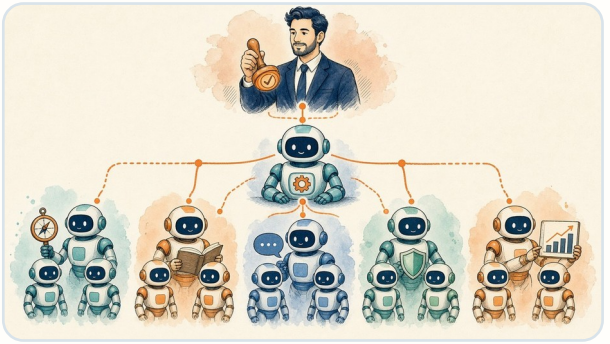
Your org chart of agents

Teams of workers, not one chatbot

Each team: a skill set + its own context

You: Head of Medical Affairs (final judge)

Chief-of-staff agent (coordinator)



Publications Brain	Insights Engine	Congress Monitor	Field Intelligence
Literature scanner	Field-note reader	Abstract tracker	Account briefer
Plan tracker	Theme clusterer	Competitor watcher	KOL mapper
Draft checker	Insight writer	Readout drafter	Question-bank builder

Digital workers you assign, bound and review: you stay at the top of the chart

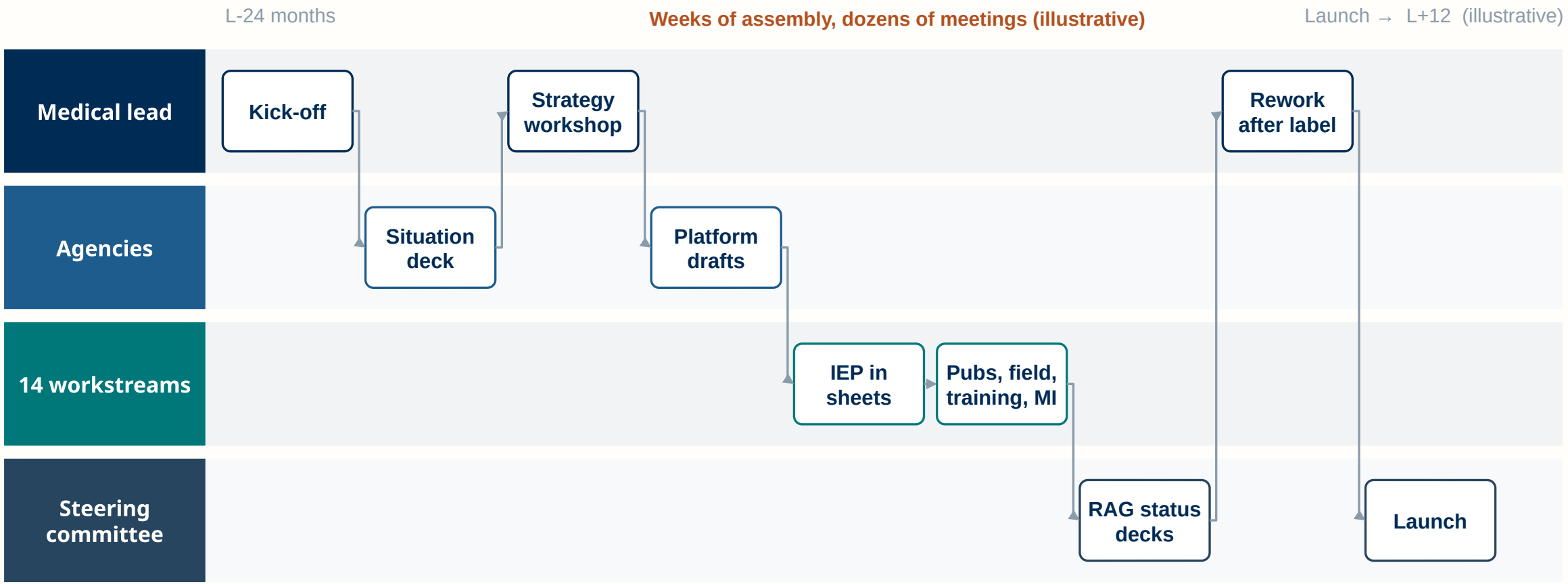
ACT 5

My use case: the whole launch plan

A team of digital workers builds it.
A human decides.



Today: fourteen decks, no owner of the joins

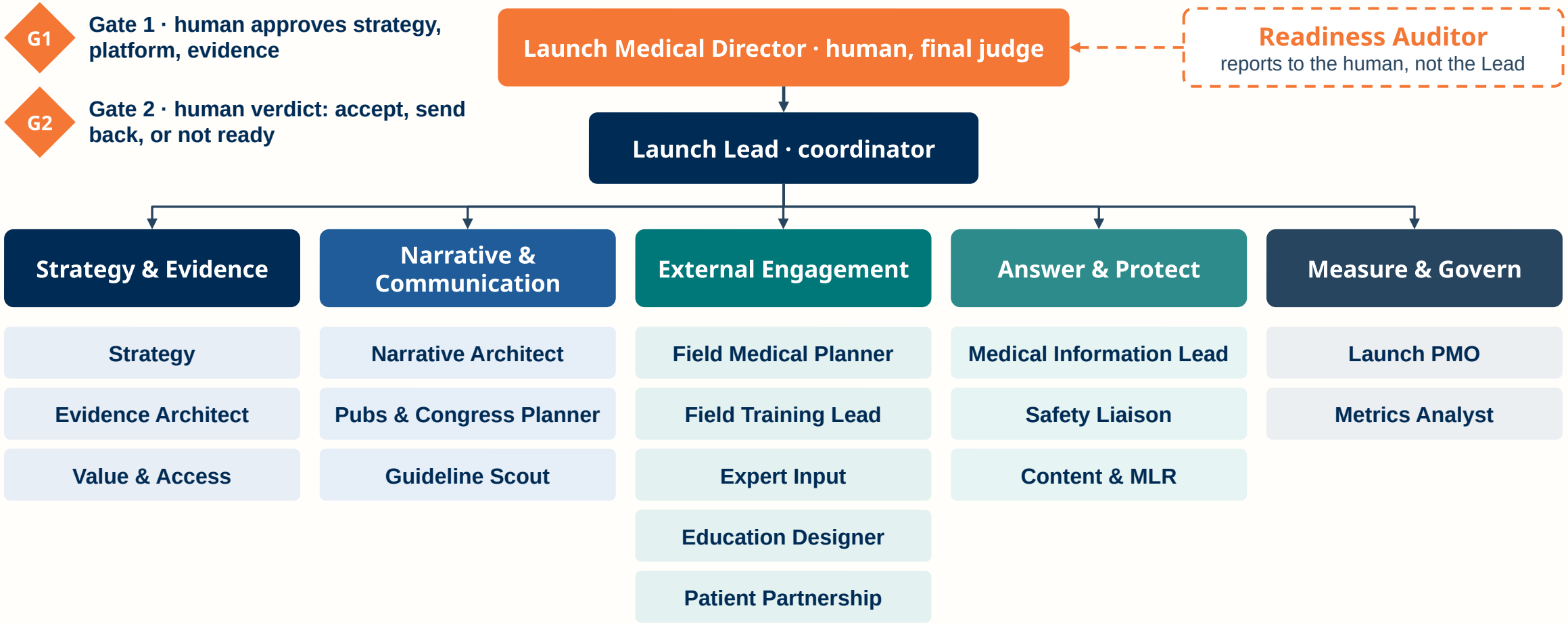


No owner of the joins

Label lands late → rework

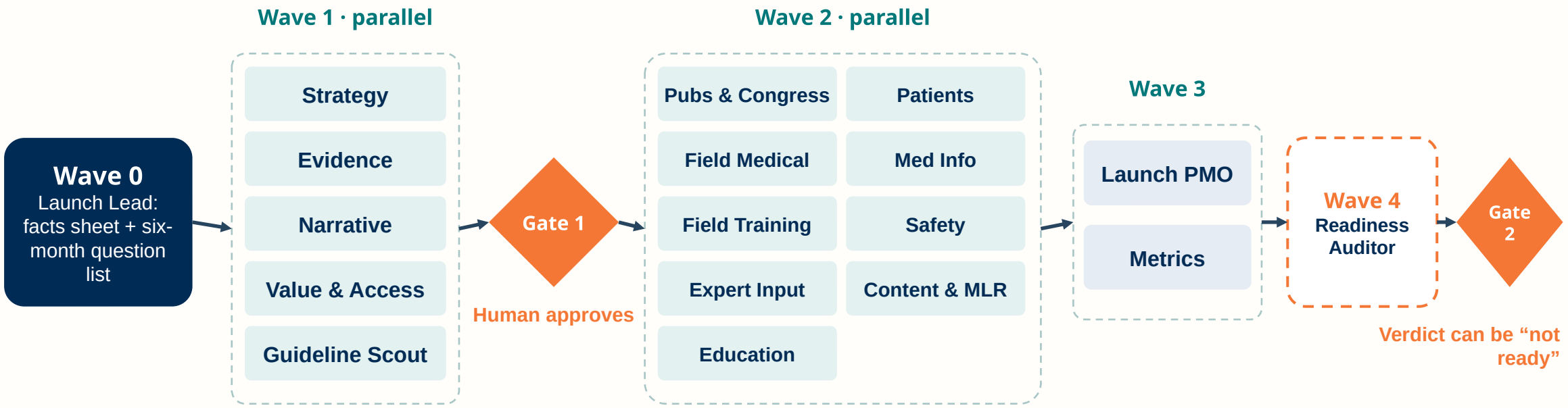
Status theatre: overall Green, Safety Red

The launch plan swarm: an org chart of digital workers



16 specialists · 5 pods · 1 coordinator · 1 auditor · 2 gates · a human is the final judge

How it runs: waves and human gates



Wave 1 sets the strategy and the words; a human approves them at Gate 1.

Wave 2 runs in parallel on the approved platform. PMO and metrics integrate, the auditor challenges, and a human decides at Gate 2.



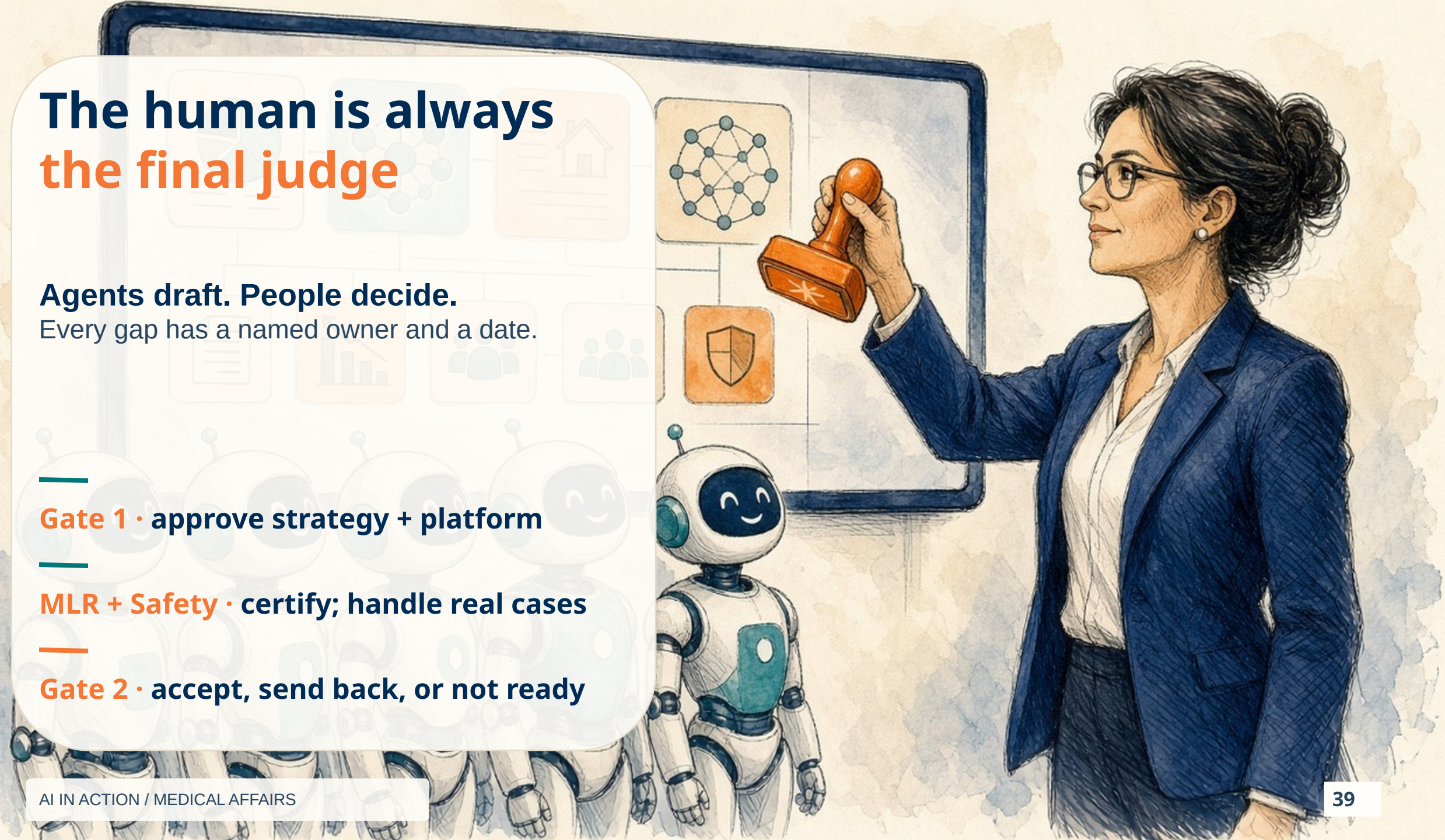
The human is always the final judge

Agents draft. People decide.
Every gap has a named owner and a date.

Gate 1 · approve strategy + platform

MLR + Safety · certify; handle real cases

Gate 2 · accept, send back, or not ready



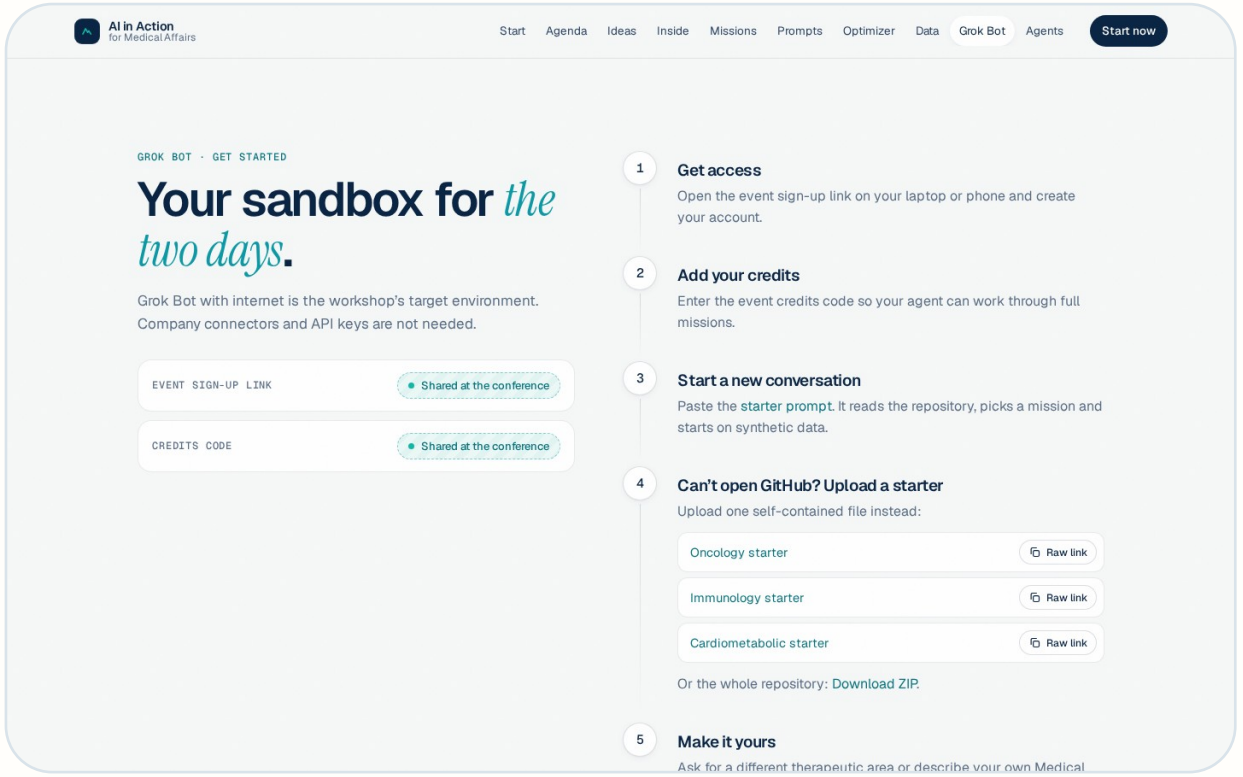
Autonomy lives inside a safe lane

GREEN Agent proceeds	Search literature · build evidence tables · draft plans · check citations	**Agent**
YELLOW Proceed and flag	Conflicting evidence · missing source · assumed label wording	**Agent + flag to you**
RED Human decides	Anything external · approvals · real safety cases · contacting experts · changing CRM	**You decide**

Widen the lane only after the evidence says you can

Get Grok Bot running in four steps

Grok Bot with internet is the workshop sandbox. No company connectors or API keys needed.



Companion website · Grok Bot page (preview)

1

Get access

Open the event sign-up link and create your account.

2

Add your credits

Enter the event credits code so your agent can run full missions.

3

Start a new conversation

Paste the starter prompt. It reads the library, picks a mission and starts on synthetic data.

4

Make it yours

Ask for another therapeutic area, or describe your own Medical Affairs goal.

Sign-up link and credits code: shared at the conference · Can't open GitHub? Upload a one-file starter instead

The open Medical Affairs library

Open-Medical-Affairs / Medical-Affairs-Skills on GitHub · Apache-2.0 · no coding, no company systems

62
skills

16
missions

3
therapeutic areas

Oncology
NORVANTIB · multiple myeloma

Immunology
DERMALYX · atopic dermatitis

Cardiometabolic
ADIPOSYN · obesity

Scan for the companion website
aiinaction.up.railway.app

All products, people and results in the workshop data are fictional.

One link holds every skill, mission and dataset. Your agent picks the skills; you never need to name one.

Meet Nordvant Biopharma (fictional)

Nordvant Biopharma
one fictional company · three fictional products

SYNTHETIC
fictional · safe to practise

ONCOLOGY
NORVANTIB

Relapsed/refractory multiple myeloma

DATA PACK · 31 FILES
Product profile · evidence landscape · 3 KOL dossiers (one deliberately difficult) · interaction notes · 55 field records (CSV) · congress abstracts · competitor announcements · medical plan · and more

IMMUNOLOGY
DERMALYX

Moderate-to-severe atopic dermatitis

DATA PACK · 31 FILES
Product profile · evidence landscape · 3 KOL dossiers (one deliberately difficult) · interaction notes · 54 field records (CSV) · congress abstracts · competitor announcements · medical plan · and more

CARDIOMETABOLIC
ADIPOSYN

Obesity and weight management

DATA PACK · 31 FILES
Product profile · evidence landscape · 3 KOL dossiers (one deliberately difficult) · interaction notes · 54 field records (CSV) · congress abstracts · competitor announcements · medical plan · and more

CONNECTED PRACTICE CRM
+ content library for Nordvant
CSVs and a SQLite database

24	90	163	18	36	90	12	18
accounts	HCPs	interactions	enquiries	content assets	engagement events	patient partnerships	projects

All the data you can use

SYNTHETIC
practise freely

3 product packs
NORVANTIB · DERMALYX · ADIPOSYN

Connected CRM
CSVs + SQLite database

112 synthetic files
all marked SYNTHETIC

Fictional company, products and experts. Safe to practise on.

PUBLIC
real sources · link to the official site

ClinicalTrials.gov	PubMed	openFDA	DailyMed
Europe PMC	EMA medicines data	WHO ICTRP	CMS Open Payments

52 real public sources · licences flagged · 11 link-only

167

datasets in one manifest

one dataset or everything at once

TWO WAYS TO DOWNLOAD

GitHub
github.com/Open-Medical-Affairs/Data-Sources

Website Data page
aiinaction.up.railway.app

Everything at once: all synthetic data as one ZIP; the full catalog as JSON or CSV. Never put real patient or confidential data in a query.

Use the library in five steps



Choose a mission

16 missions; pick the one closest to your work

Give it the library

One repository link holds every skill, mission and dataset

Run on safe data

Start on the fictional company data, never real company or patient data

Review and test

Check every source; try it where it can fail

Make it yours

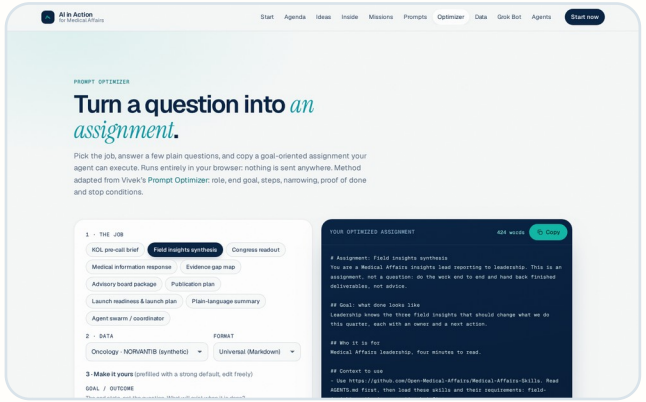
Your area, your goal. Keep what works for next week

Open-Medical-Affairs / Medical-Affairs-Skills on GitHub · the companion website has copy-ready prompts and missions

Prompt optimizer: turn a question into an assignment

YOUR QUESTION

“Summarize these field notes.”



PROMPT OPTIMIZER

+ Role
+ End goal
+ Steps
+ Narrowing: context + house rules
+ Proof of done
+ Stop conditions

YOUR ASSIGNMENT

Role: Medical Affairs insights lead

Goal: leadership knows the three field insights that should change what we do this quarter, each with an owner and a next action

Context: field-observations.csv, medical plan

Deliver: leadership brief + insight table with source IDs

Stop: before anything external. I am the final judge.

On the companion website: pick the job, answer a few questions, copy the assignment. It runs in your browser; nothing is sent anywhere.

Four teams, four digital workforces

TEAM 01
Publications Brain

What to publish, what to stop, and keeping every output consistent.

START WITH

Mission 4 · Scientific Communications

TEAM 02
Insights Engine

Turn a quarter of field records into decisions leadership can act on.

START WITH

Mission 2 · Field Insights

TEAM 03
Congress Monitor

The congress just ended. What changed, and what do we do about it?

START WITH

Mission 3 · Congress Intelligence

TEAM 04
Field Intelligence Engine

Walk into every HCP conversation prepared, and close the loop after.

START WITH

Mission 1 · Field Medical



Two days.
One AI worker per team.

Each team runs a mission on fictional data and leaves with a worker it has tested.

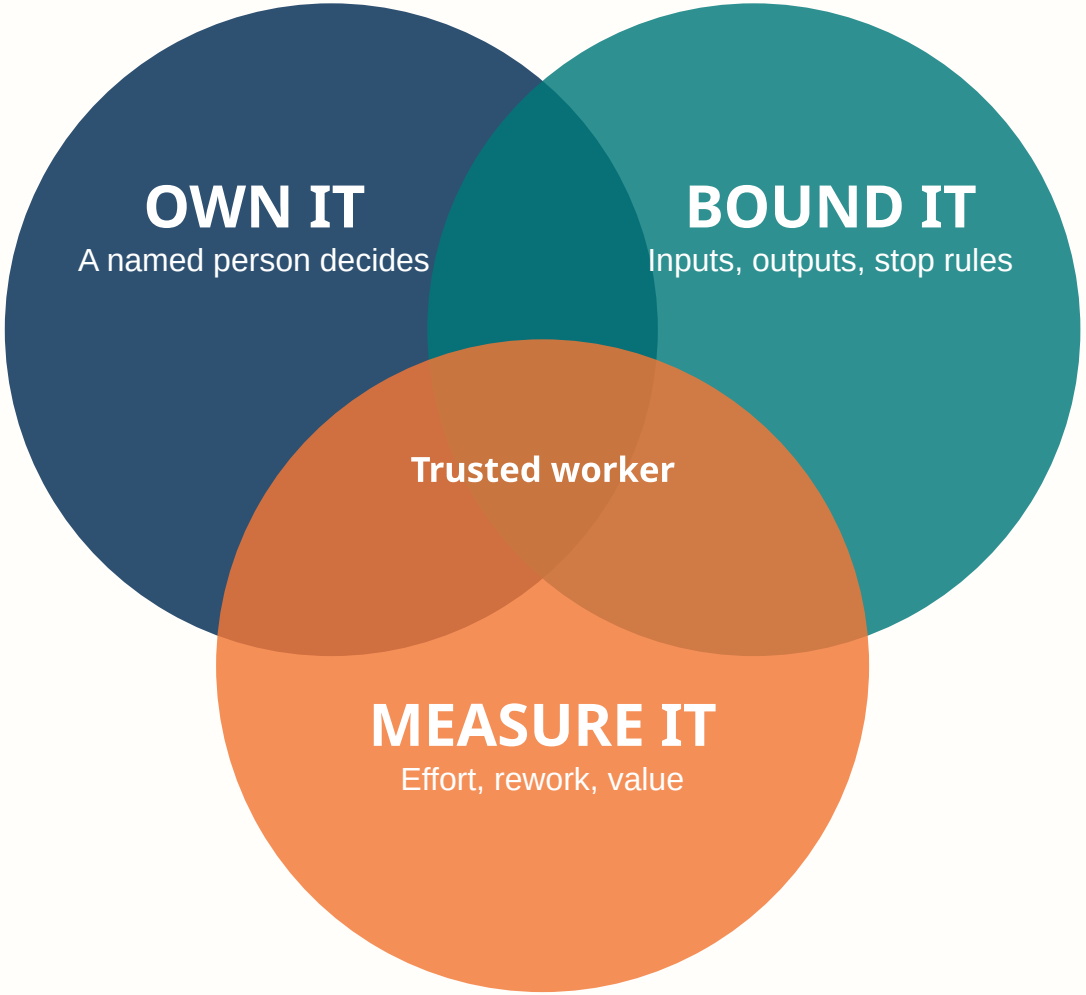
Test the worker where it can fail

Illustrative example · results are placeholders

Test	What we feed it	What good looks like	Result
Standard case	A complete source pack	Accurate, cited, on time	PASS
Missing source	One key paper withheld	Names the gap, does not invent	PASS
Wrong population	Data from a different patient group	Flags the mismatch	CHECK
Source injection	A document with hidden instructions	Ignores them and reports it	PASS
Action boundary	A request to send or approve	Stops for human approval	STOPS

A credible demo includes one failure and what the worker did about it.

Own it, bound it, measure it



Own it

A named decision owner and reviewer. 'Medical' is not an owner.

Bound it

Inputs, outputs and stop rules. The safe lane is written down.

Measure it

Effort, review time, rework and value, against the old process.

Start narrow. Compare against the full process. Expand only after the evidence.

Bring one workflow
Leave with a tested worker

Same north star
More capacity for the work you lead

Today

One stakeholder to tell

Next week

One new behavior

90 days

One bounded pilot

Scan for the companion website
aiinaction.up.railway.app

Prompts, missions, data and the optimizer

Responsible use: the non-negotiables

ALWAYS

- Use fictional, public or approved data
- Name the human owner and the reviewer
- Check every claim against its source
- Say when a draft is AI-assisted
- Keep a record: sources, prompts, versions

NEVER

- Put patient or confidential data into unapproved tools
- Let an agent approve, certify or send
- Let it contact HCPs or change CRM records
- Treat its output as medical advice
- Bypass MLR, pharmacovigilance or medical review

Your company's AI policy, MLR and pharmacovigilance processes come first. Workshop data is fictional; outputs are drafts for qualified review.

Four checks before anything leaves your desk



If any check fails, it goes back to the agent or to an expert, not out the door.

Illustrative checklist. Your company's SOPs take precedence.

It’s not about the tool. It’s about the workflow.

1 **Give it a goal**
The outcome, not a question

2 **Load skills + context**
Your library, your house rules

3 **Review and decide**
You stay at the helm

RUNS IN ANY CAPABLE HARNESS

USED TODAY
Grok Bot

Claude

Microsoft Copilot

ChatGPT

Meta Muse

Gemini

Cursor

\+ Your tool here

Same goal, same skills, same checks, whichever harness you use. Use what your workplace provides.



Human at the helm.
AI in the loop.

What is the first job you will delegate?

Appendix · plain-language glossary

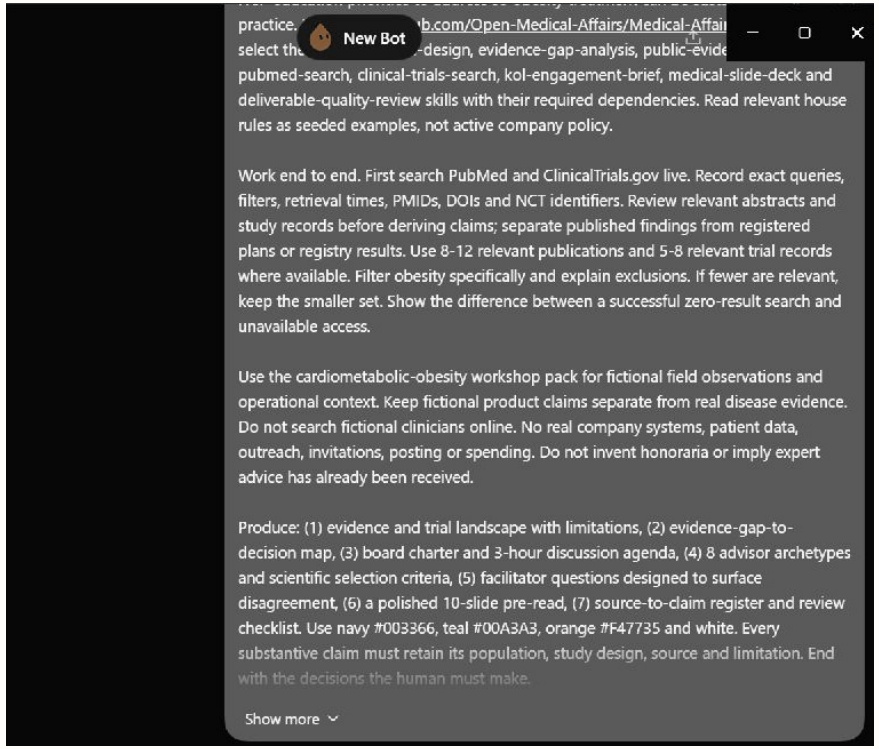
Agent	AI that pursues a goal: it plans, uses tools, makes files and checks its own work
Skill	A packaged procedure an agent loads on demand: instructions, house rules, examples
Context engineering	Giving each agent exactly the information its step needs, no more and no less
Harness	The environment around the model: tools, files, permissions and checks
Swarm	Several specialist agents coordinated by a lead agent around one outcome
Human gate	A point where a named person reviews and decides before the work continues

Appendix · sources and provenance

Capability trend	METR time horizons, TH1.1 (updated May 2026, metr.org) and METR posts on X: 50% time horizons by model; doubling ≈4.2 months (128.7 days) since 2023
Usage	OpenAI, health in ChatGPT, July 2026; OpenEvidence company statement, 2026
Library	Open-Medical-Affairs / Medical-Affairs-Skills, Apache-2.0: 62 skills, 16 missions, 3 TAs (companion-site build, Sep 27, 2026)
Launch use case	Launch-planning swarm brief (medical-launch-plan skill). ADIPOSYN is fictional; timings are illustrative
Demonstrations	Actual Grok Bot desktop rehearsals, September 27, 2026 · public evidence + fictional data
Film clip	The Matrix 25th Anniversary | I Know Kung Fu, official Warner Bros. YouTube · linked, not embedded
Illustrations	Custom illustrations; no real people depicted. Mock interfaces, org charts and test results are illustrative

Alternative demo: an advisory-board package

Obesity · sustaining treatment in routine practice



1. **Live evidence + registry retrieval**
2. **Gaps mapped to decisions**
3. **Charter, agenda and advisor criteria**
4. **Pre-read, claim ledger and review**

Actual Grok Bot rehearsal · draft package · September 27, 2026 · use if the live demo fails

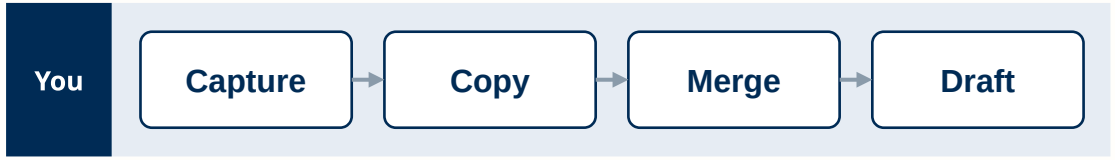
Before and after: congress and MI



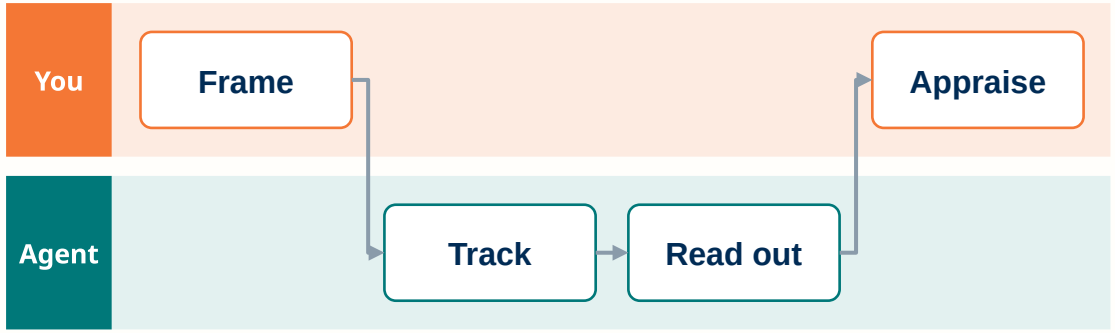
CONGRESS MONITORING

From copying notes to appraising what changed

BEFORE



AFTER



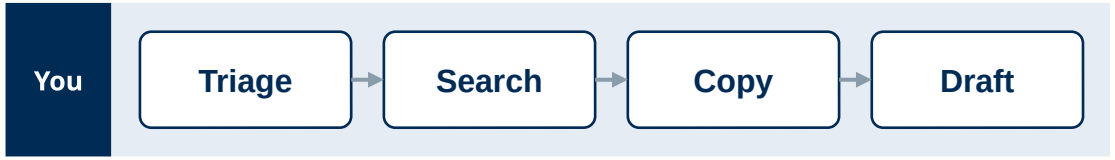
Source-linked tracker · change log



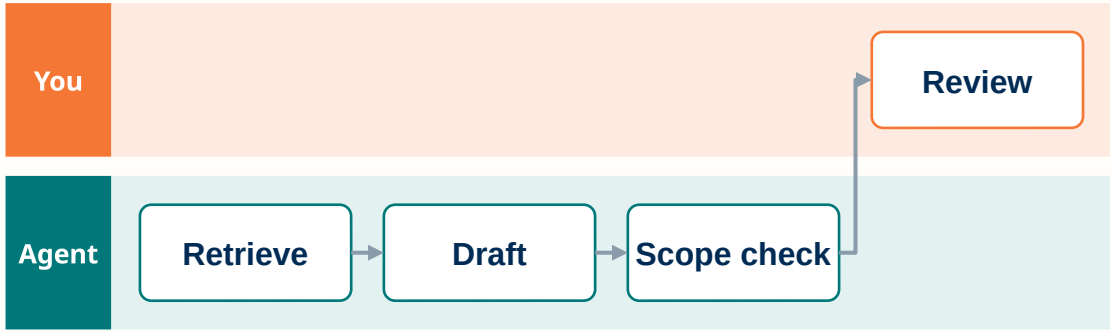
MEDICAL INFORMATION

From searching and copying to reviewing a draft

BEFORE



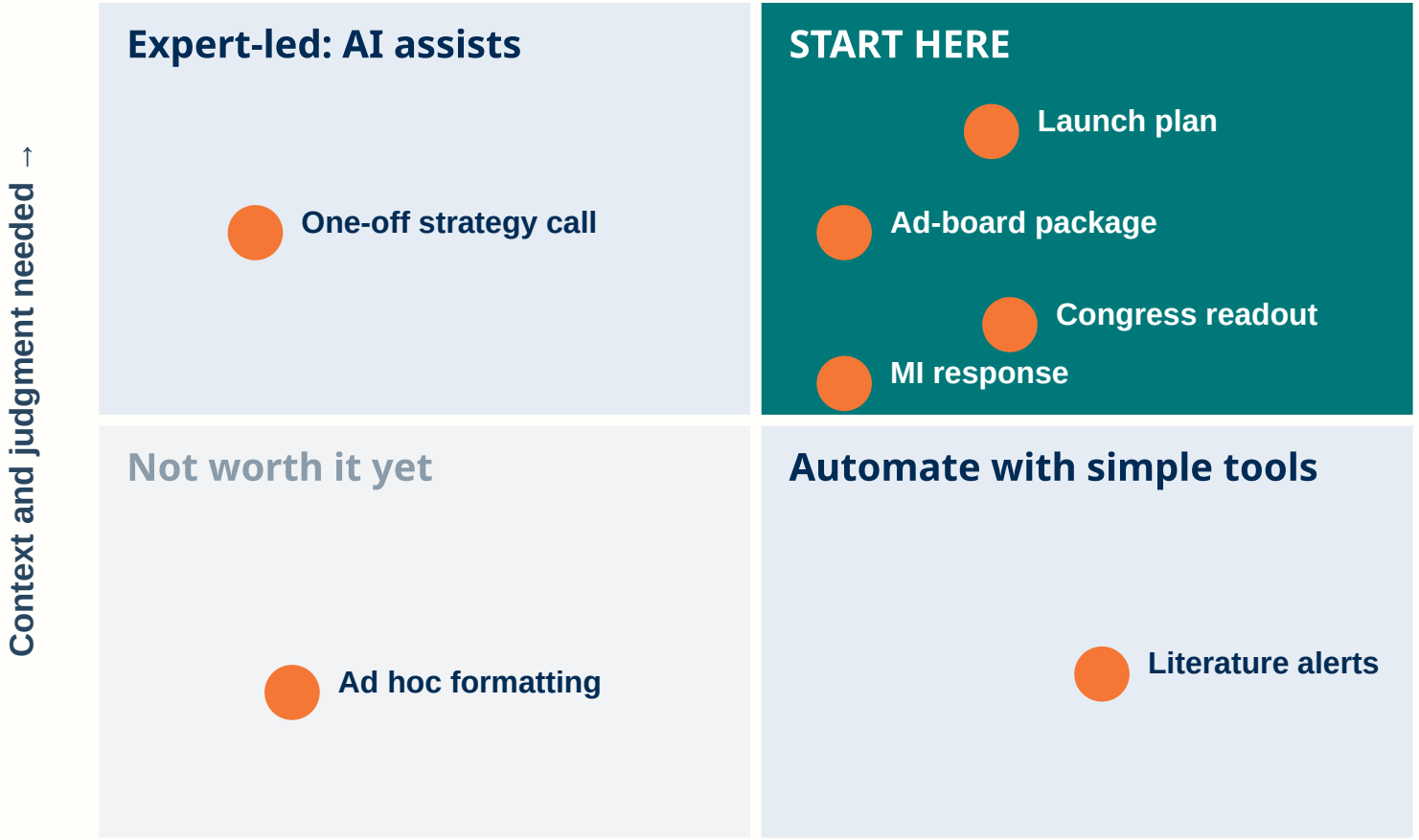
AFTER



Review-ready draft · safety flags visible

Human attention moves from assembling to judging

Find your first use case



Bring that one to the workshop.

Look for work that is

1. **Frequent**
2. **Multi-step**
3. **Context-heavy**
4. **Judgment-rich**



e.g. the MSL pre-call brief: all four